GAZE: Grounding-Aware Zone-Adaptive Visual Token Selection with Focal–Spatial Synergy for Efficient MLLMs

Supplementary Appendix

A Benchmarks and Model Descriptions

A.1 Benchmark Descriptions

We evaluate our method on 10 widely adopted multimodal benchmarks, spanning two document and chart understanding benchmarks, one scene text recognition benchmark, two science and knowledge reasoning benchmarks, three comprehensive perception and reasoning benchmarks, and two general visual question answering benchmarks. All inference settings and evaluation metrics follow the original benchmark configurations.

DocVQA [16] is a document visual question answering benchmark constructed from real scanned industrial documents, including invoices, tables, letters, and forms. Models are required to answer natural language questions based on image content. Performance is measured by Average Normalized Levenshtein Similarity (ANLS), which tolerates minor OCR errors. This benchmark demands precise text localization and reading comprehension, making it a key indicator of visual token preservation quality.

ChartQA [15] consists of bar charts, line charts, and pie charts sourced from real news articles and statistical reports. Questions are either human-authored or machine-generated, requiring models to reason about numerical values, trends, and comparative relationships. Performance is measured by Relaxed Accuracy, which accepts numerical answers within a 5% margin of error. ChartQA is sensitive to visual token distributions over non-uniform images.

OCRBench [13] is a comprehensive OCR evaluation benchmark covering 29 sub-tasks, including scene text recognition, handwritten text recognition, document text extraction, and table structure reconstruction, with a total of 1,000 questions. It systematically evaluates model performance across diverse OCR scenarios, with scores reported as percentages.

ScienceQA [14] is a multimodal science question answering dataset covering natural science, social science, and language science topics from elementary to high school levels. A subset of questions is paired with images. Each question is multiple-choice and accompanied by explanations and annotations. Since many questions can be answered without images, this benchmark measures model robustness in weakly vision-dependent scenarios.

AI2D [8] is a diagram understanding dataset from the Allen Institute for AI, containing over 5,000 grade school science diagrams paired with more than 15,000 multiple-choice questions such as food chains, Earth structure illustrations, and cell division processes. Questions require understanding of arrows, labels, and structural relationships, placing high demands on spatial semantic comprehension.

MMMU [18] is a large-scale multidisciplinary multimodal understanding and reasoning benchmark spanning six domains: art, business, science, health, humanities, and technology and engineering. It comprises 11,500 questions across 30 subjects, with image types including charts, tables, musical scores, and chemical formulas. MMMU requires both domain-specific knowledge and multi-step reasoning, making it one of the most challenging multimodal benchmarks. Performance is measured by overall accuracy.

MME [5] is a comprehensive evaluation benchmark designed to assess the multi-dimensional capabilities of MLLMs, covering 14 sub-tasks across two categories: perception and cognition. Subtasks include existence judgment, counting, position understanding, color recognition, OCR, commonsense reasoning, mathematical reasoning, and code reasoning. Each question is a binary yes/no judgment. We report the sum of perception and cognition scores.

POPE [10] is a benchmark designed to evaluate object hallucination in MLLMs. It measures hallucination by asking models whether a specific object exists in an image. Performance is measured by F1 score. The dataset includes three sampling strategies: random, popular, and adversarial, with increasing levels of difficulty. We report the average F1 score across all three strategies.

VizWiz [6] is a visual question answering dataset collected from real blind users via a mobile application. Images vary significantly in quality and include blur, occlusion, and skewed angles. Some questions are objectively unanswerable due to insufficient or ambiguous visual information. VizWiz challenges model robustness on low-quality, real-world images, and performance is measured by VQA accuracy.

MMB^{CN} [12] is the Chinese version of MMBench, covering approximately 20 fine-grained capability dimensions including attribute recognition, relational reasoning, and logical reasoning. All questions are posed in Chinese, directly evaluating models on Chinese multimodal understanding. It also serves as a testbed for assessing cross-lingual transferability of multimodal capabilities. Performance is measured by accuracy.

A.2 Model Architectures

To cover diverse architectural designs and resolution strategies, we evaluate GAZE on four representative vision-language models.

LLaVA-1.5-7B [11] is one of the most widely adopted open-source vision-language models. LLaVA-1.5 uses a pretrained CLIP ViT-L/14 as the visual encoder and Vicuna-7B as the text decoder, connected via a two-layer MLP. Compared to the original LLaVA, the input resolution is increased from 224 to 336, producing a fixed set of 576 visual tokens. The fixed token count and non-tiling design provide a controlled setting for evaluating GAZE pruning under a simple architecture.

LLaVA-OneVision-1.5-8B [9] replaces the visual encoder with the more powerful SigLIP SO400M and retains the two-layer MLP as the vision-language projection layer. The language backbone is upgraded to Qwen2-7B, yielding substantial improvements in instruction following and reasoning capabilities. The model adopts the AnyRes dynamic high-resolution scheme, which splits input

Table 1: Performance comparison of visual token pruning methods on Qwen2.5-VL-3B. Our GAZE consistently achieves the best trade-off between efficiency and accuracy across all ratios. Avg (%) is the mean ratio relative to the full-token baseline (100%). * denotes methods involving additional training.

Method	DocVQA Anls	ChartQA Relaxed	OCRBench Acc	ScienceQA EM	AI2D EM	MMMU Acc	MME Score	POPE F1	Avg
Dynamic Resolution (MinPix=256×28×28, MaxPix=2048×28×28), Upper Bound (100%)									
Qwen2.5-VL-3B	92.76	83.40	78.80	80.37	90.84	46.78	2168.46	86.94	100%
Retain 30% Tokens (70% Compression Ratio)									
FastV [3] (ECCV'24)	81.66	70.04	62.90	78.63	86.40	46.00	2048.53	82.94	91.63
PruMerge+ [17] (ICCV'25)	65.43	63.88	55.30	78.93	79.24	46.00	2051.01	84.78	86.65
DivPrune [1] (CVPR'25)	73.99	66.48	63.10	78.18	83.42	45.22	2010.72	86.38	89.68
PruneSiD [4] (ICLR'26)	72.91	70.44	55.40	79.67	76.78	45.33	1978.81	85.78	87.98
Dynamic* [7] (ICLR'25)	85.28	75.40	73.80	79.67	87.82	45.22	2078.06	85.13	95.28
VisionSelector* [19] (arXiv'25)	88.83	73.24	75.20	80.07	88.50	**46.11**	2085.46	83.79	95.90
GAZE (Ours)*	**90.15**	**78.36**	**75.80**	**80.42**	**88.96**	45.89	**2086.27**	**87.66**	**97.54**
Retain 20% Tokens (80% Compression Ratio)									
FastV [3] (ECCV'24)	72.90	62.00	56.00	78.88	83.29	45.11	1992.58	80.01	86.77
PruMerge+ [17] (ICCV'25)	52.58	54.76	48.40	79.97	79.74	44.89	1977.90	83.23	81.74
DivPrune [1] (CVPR'25)	63.82	57.64	53.50	**80.46**	77.59	45.56	1955.56	**85.47**	84.65
PruneSiD [4] (ICLR'26)	62.51	56.96	46.70	79.82	72.34	44.56	1901.86	85.40	81.89
Dynamic* [7] (ICLR'25)	76.47	70.20	67.20	78.83	85.49	44.33	**2053.26**	83.34	91.17
VisionSelector* [19] (arXiv'25)	83.63	70.04	69.90	79.62	86.59	45.89	2045.90	82.09	93.01
GAZE (Ours)*	**86.42**	**76.00**	**73.50**	79.87	**87.40**	**45.89**	2038.32	85.45	**95.43**
Retain 10% Tokens (90% Compression Ratio)									
FastV [3] (ECCV'24)	54.62	46.96	41.10	79.33	77.07	45.11	1815.70	70.28	76.49
PruMerge+ [17] (ICCV'25)	37.51	46.68	35.00	78.93	68.43	44.56	1777.60	77.55	72.60
DivPrune [1] (CVPR'25)	47.13	44.36	39.40	78.19	70.14	44.67	1782.74	80.50	74.85
PruneSiD [4] (ICLR'26)	43.23	45.28	37.30	77.94	67.45	44.56	1830.72	80.26	73.93
Dynamic* [7] (ICLR'25)	58.28	61.56	55.60	79.08	**86.66**	44.33	1887.79	75.01	83.64
VisionSelector* [19] (arXiv'25)	67.79	61.64	56.70	79.62	81.09	44.67	1928.75	77.03	85.04
GAZE (Ours)*	**75.85**	**68.00**	**62.50**	**80.37**	82.58	**45.56**	**1945.97**	**82.23**	**89.41**

images into multiple sub-images based on resolution and encodes them independently, resulting in a variable number of visual tokens.

Qwen2.5-VL-7B-Instruct [2] contains a ViT-based visual encoder and a 7B language decoder, connected by an MLP-based vision-language merger that compresses visual token sequences by grouping spatially adjacent patch features and projecting them into the language embedding space. The model adopts a dynamic resolution tiling scheme that automatically partitions input images and encodes each tile independently, producing a variable number of visual tokens.

Qwen2.5-VL-3B-Instruct shares the same architectural design as the 7B variant, including the ViT-based visual encoder, MLP-based vision-language merger, and dynamic resolution tiling scheme, with the language decoder reduced to 3B parameters. This model is used to evaluate the pruning effectiveness and performance retention of GAZE in lightweight deployment scenarios.

B Additional Experimental Results

B.1 Results on Qwen2.5-VL-3B

To verify the generalization ability of GAZE on lightweight models, we conduct a systematic comparison with existing training-based and training-free methods on Qwen2.5-VL-3B across eight benchmarks: DocVQA, ChartQA, OCRBench, ScienceQA, AI2D, MMMU, MME, and POPE. The results are reported in Table 1. GAZE achieves the highest average retention rate under all three token budgets, and its advantage over the second-best method grows steadily as

Table 2: Performance comparison of visual token pruning methods on LLaVA-1.5-7B (fixed 336×336 resolution, 576 tokens). GAZE achieves the highest average accuracy retention across all token budgets, demonstrating strong cross-architecture generalizability.

Method	VizWiz Acc	SQA^I Acc	POPE F1	MME Score	MMB^{CN} Acc	Avg
Fixed Resolution (576 Tokens), Upper Bound (100%)						
LLaVA-1.5-7B	50.1	69.5	85.9	1862.0	58.1	100%
Retain 128 Tokens (78% Compression Ratio)						
FastV [3] (ECCV'24)	50.5	59.6	62.8	1088.4	48.8	80.4
PruMerge+ [17] (ICCV'25)	53.7	**69.1**	83.1	1554.0	**55.8**	96.6
DivPrune [1] (CVPR'25)	52.8	68.6	85.5	1707.0	52.3	97.1
GAZE (Ours)	**55.9**	68.9	**86.2**	**1727.2**	52.4	**98.8**
Retain 64 Tokens (89% Compression Ratio)						
FastV [3] (ECCV'24)	50.8	51.1	48.0	1019.6	42.7	71.8
PruMerge+ [17] (ICCV'25)	53.7	**69.5**	75.7	1640.0	52.1	94.6
DivPrune [1] (CVPR'25)	53.6	68.0	**85.5**	1615.0	**52.3**	96.2
GAZE (Ours)	**56.5**	68.7	85.0	**1708.6**	49.7	**97.6**
Retain 32 Tokens (94% Compression Ratio)						
FastV [3] (ECCV'24)	51.7	42.6	32.5	884.6	33.2	61.4
PruMerge+ [17] (ICCV'25)	53.5	67.9	66.7	1550.4	45.9	88.9
DivPrune [1] (CVPR'25)	53.3	**68.6**	81.5	**1594.3**	**49.1**	94.0
GAZE (Ours)	**55.6**	68.5	**81.9**	1587.9	47.9	**94.5**

the compression ratio increases, from 1.64 percentage points at the 30% budget to 4.37 percentage points at the 10% budget. This confirms that GAZE exhibits superior robustness under aggressive compression. Notably, at the 30% budget, GAZE surpasses the full-token baseline on POPE, indicating that moderate visual token pruning can improve the model's ability to focus on critical visual cues by removing redundant tokens, thereby achieving lossless or even beneficial compression. Overall, GAZE attains strong performance on the 3B-scale model with low training overhead, further validating its generalization across model scales.

B.2 Results on LLaVA-1.5-7B

To evaluate the performance of GAZE on fixed-resolution architectures, we benchmark it on LLaVA-1.5-7B (576 tokens), as shown in Table 2. Unlike the dynamic-resolution Qwen2.5-VL models used in the main text, LLaVA-1.5-7B adopts a fixed 336×336 input resolution, yielding shorter token sequences with notably different internal feature distributions compared to the Qwen series. This setup thus serves as an effective test bed for assessing the cross-architecture transferability of GAZE. Since this model exhibits limited baseline performance on document understanding and OCR tasks that are sensitive to high resolution, we select five benchmarks centered on general visual question answering and perception: VizWiz, SQA^I, POPE, MME, and MMB^{CN}. We evaluate under three absolute token budgets of 128, 64, and 32.

Across the 128, 64, and 32 token budgets, GAZE consistently achieves the highest average retention rate. Notably, at the 128-token budget, GAZE obtains 86.2 on POPE, surpassing the full-token baseline, once again confirming that moderate visual token compression can improve the model's discriminability of critical visual cues by removing redundant and distracting tokens, thereby yielding gain-positive compression. Under the extreme 32-token compression scenario, FastV degrades to 61.4%, whereas GAZE maintains 94.5%, leading by 33.1 percentage points. These results collectively demonstrate that GAZE exhibits significant advantages on fixed-resolution architectures as well, validating its strong generalization across diverse architectural designs.

Table 3: Generalization across retention budgets on Qwen2.5-VL-7B-Instruct. Models are trained with either a fixed retention ratio k or GAZE's dynamic bbox-based budget, then evaluated at different test-time retention ratios.

Method	ChartQA Relaxed	OCRBench Acc	POPE F1	MME Score	Avg
Dynamic Resolution (MinPix=256×28×28, MaxPix=2048×28×28), Upper Bound (100%)					
Qwen2.5-VL-7B	83.40	83.80	86.19	2342.15	100%
Retain 10% Tokens (90% Compression Ratio)					
Fixed k=0.1	58.92	67.10	**82.82**	2157.62	84.73
Fixed k=0.2	61.68	**69.80**	82.23	2215.46	86.81
Fixed k=0.3	60.88	68.80	82.06	2197.00	86.03
GAZE (Ours)	**62.04**	69.10	82.58	**2222.20**	**86.88**
Retain 20% Tokens (80% Compression Ratio)					
Fixed k=0.1	69.84	78.70	**84.91**	2316.61	93.77
Fixed k=0.2	**71.20**	78.00	84.83	2313.65	93.91
Fixed k=0.3	70.56	78.10	84.84	2296.78	93.58
GAZE (Ours)	71.12	**79.20**	84.53	**2339.79**	**94.44**
Retain 30% Tokens (70% Compression Ratio)					
Fixed k=0.1	74.56	81.40	85.63	2331.16	96.35
Fixed k=0.2	74.84	81.10	85.30	**2345.77**	96.41
Fixed k=0.3	**76.12**	81.20	**85.81**	2334.04	96.85
GAZE (Ours)	75.92	**82.00**	85.48	2338.60	**96.98**

C Hyperparameter Sensitivity Analysis

C.1 Generalization Across Retention Budgets

To verify whether the dynamic budget constructed from bbox area ratios generalizes across retention rates, we compare GAZE with three fixed-ratio training schemes (Fixed $k \in \{0.1, 0.2, 0.3\}$) on Qwen2.5-VL-7B-Instruct. Results are reported in Table 3. GAZE achieves the highest average retention rate under all three test-time budgets, making it the only scheme that ranks first at every budget setting. In contrast, fixed-ratio schemes occasionally outperform GAZE on individual benchmarks, but such advantages are neither stable nor consistent: the best-performing fixed scheme varies across test conditions and tasks, and does not always correspond to the ratio that matches the test-time budget. This indicates that the scattered per-benchmark gains cannot be simply attributed to train-test ratio alignment. Furthermore, at the 10% budget, Fixed k=0.1, which exactly matches the test condition, yields the lowest average score among all schemes, suggesting that an excessively low fixed ratio degrades visual representation quality during training. Moreover, at the 20% budget, GAZE achieves an MME score of 2339.79, only 0.10% below the full-token baseline, showing that the token selection strategy learned via dynamic budget training preserves the model's comprehensive perception capability in a

Table 4: Hyperparameter sensitivity analysis of the consistency constraint schedule weight λ_{end} and focal channel budget ratio k_{foc} on Qwen2.5-VL-7B at 10% token retention.

Cfg.	λ_{start}	λ_{end}	k_{foc}	ChartQA	OCRBench	AI2D	MME	POPE	Avg
1	0.1	1.0	0.9	59.56	66.20	85.40	2169.18	81.10	85.68
2	0.1	3.0	0.9	58.60	67.10	86.30	2161.40	80.98	85.76
3	0.1	2.0	0.8	57.60	66.50	83.35	2155.89	81.83	84.89
4	0.1	2.0	0.7	61.56	67.20	84.23	2167.87	82.21	86.39
5	0.1	3.0	0.7	61.40	69.10	85.91	**2227.54**	81.27	87.46
GAZE	0.1	2.0	0.9	**62.04**	**69.10**	**86.37**	2222.20	**82.58**	**87.97**

nearly lossless manner even under moderate compression. Overall, no single fixed-ratio scheme can consistently outperform GAZE across different retention rates. In contrast, GAZE naturally encounters a wide range of retention ratios matched to the information density of each training image, thereby learning a token selection strategy that remains robust under varying compression strengths. These results further validate the effectiveness and necessity of the proposed dynamic budget design.

C.2 Effect of λ_{end} and k_{foc}

To examine how λ_{end} and k_{foc} affect performance, we conduct a systematic ablation on Qwen2.5-VL-7B at 10% token retention. Results are reported in Table 4. All configurations fix $\lambda_{\text{start}} = 0.1$.

Effect of λ_{end}. With $k_{\text{foc}} = 0.9$, we compare $\lambda_{\text{end}} \in \{1.0, 2.0, 3.0\}$. Setting $\lambda_{\text{end}} = 1.0$ provides insufficient constraint strength, leaving the gap between the soft mask and the hard selection under-regularized. The average retention rate is only 85.68%. Increasing λ_{end} to 2.0 yields the globally best result of 87.97%. Further raising it to 3.0 causes a drop to 85.76%, indicating that an overly strong binarization regularizer prematurely shrinks the optimization freedom of the continuous scoring space and creates gradient conflicts with $\mathcal{L}_{\text{bbox}}$ when the Focal budget dominates. Under $k_{\text{foc}} = 0.7$, however, a different trend emerges: raising λ_{end} from 2.0 to 3.0 improves the average retention rate from 86.39% to 87.46%. This suggests that a larger Spatial budget benefits from a stronger consistency constraint, which helps stabilize the cooperative behavior of the two channels. These observations reveal an interaction effect between λ_{end} and k_{foc}: the two hyperparameters cannot be tuned in isolation. Nevertheless, none of the $k_{\text{foc}} = 0.7$ configurations surpass the best result achieved with $k_{\text{foc}} = 0.9$ and $\lambda_{\text{end}} = 2.0$.

Effect of k_{foc}. With $\lambda_{\text{end}} = 2.0$, we compare $k_{\text{foc}} \in \{0.7, 0.8, 0.9\}$. The highest average retention rate of 87.97% is achieved at $k_{\text{foc}} = 0.9$. Reducing it to 0.8 and 0.7 lowers the rate to 84.89% and 86.39%, respectively. Notably, $k_{\text{foc}} = 0.7$ outperforms 0.8, suggesting that once the Spatial budget reaches a sufficient scale, the grid prior can more effectively exploit its spatial coverage advantage and partially compensate for the reduced precision of the Focal channel. Under $\lambda_{\text{end}} = 3.0$, the trend differs: decreasing k_{foc} from 0.9 to 0.7 instead raises the average retention rate from 85.76% to 87.46%. This is consistent with the finding in the λ_{end} ablation above: a larger Spatial budget combined with a stronger consistency constraint can be mutually complementary in certain regimes. Yet this combination still falls short of the best result at $k_{\text{foc}} = 0.9$ and $\lambda_{\text{end}} = 2.0$. These results indicate that the budget allocation should be Focal-dominant to ensure the retention of core semantic tokens, while reserving a small Spatial budget ($k_{\text{foc}} = 0.9$ corresponds to a 10% Spatial ratio) to provide necessary spatial coverage and global context. Both channels are indispensable. In summary, the default configuration of GAZE ($\lambda_{\text{start}} = 0.1$, $\lambda_{\text{end}} = 2.0$, $k_{\text{foc}} = 0.9$) consistently achieves the best performance across all ablation combinations, validating the rationality and robustness of each hyperparameter choice.

D Qualitative Visualization

To demonstrate the advantages and limitations of the dual-channel token selection mechanism in GAZE, we visualize the retained tokens at 20% and 10% retention ratios on representative VQA samples, with FastV, DivPrune, and PruneSiD as comparison baselines.

FastV exhibits a strong spatial bias, tending to cover large low-information-density regions while neglecting semantically important areas, which leads to severe performance degradation on detail-sensitive tasks. DivPrune achieves relatively uniform spatial coverage but lacks concentration on key semantic regions, frequently missing critical objects in fine-grained recognition and counting tasks, especially when visual cues are spatially compact. PruneSiD shows limited spatial adaptability, with its retained tokens often failing to cover task-relevant regions in scenarios that require precise localization, often resulting in critical information loss.In contrast, GAZE unifies precise focus and global coverage through its Focal–Spatial dual-channel design. The Focal channel (gold pixels) leverages learned importance scores to accurately attend to semantically rich key regions, while the Spatial channel (blue pixels) supplements essential spatial context and broader global information through uniform grid sampling. As shown in Fig. 2, at 20% retention ratio, GAZE produces correct answers on the majority of tasks, consistently outperforming all baseline methods. We also present failure cases: in person counting under occlusion and fine-grained numerical comparison in charts, the model still yields incorrect predictions even though GAZE retains query-relevant visual tokens. These errors are primarily attributed to information loss caused by occlusion and insufficient feature discrimination under compression. Under the extreme compression setting of 10% retention ratio (Fig. 1), GAZE answers all five samples correctly while all baseline methods fail, validating the robustness of the dual-channel mechanism even at extremely high compression ratios.

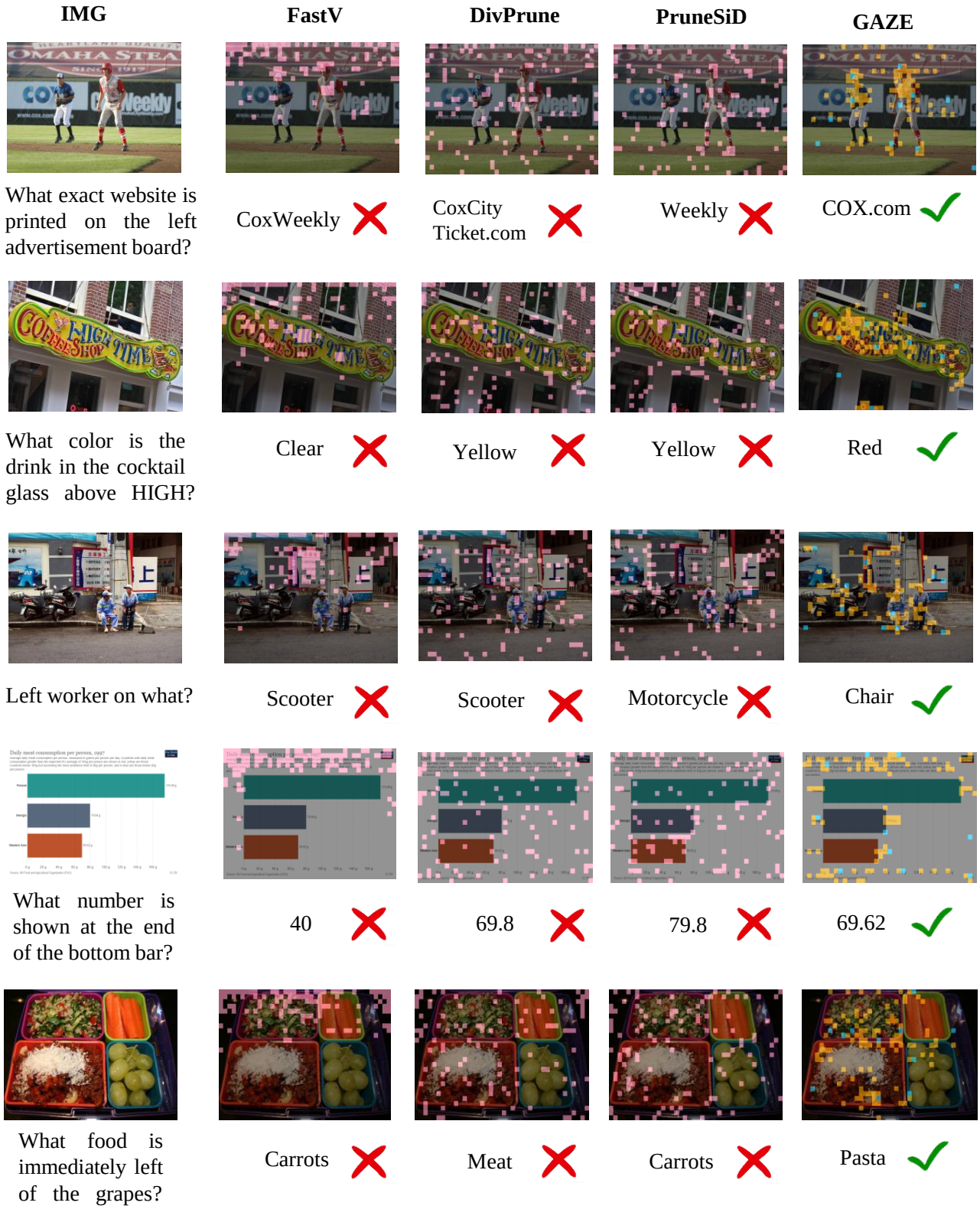


Figure 1: Qualitative comparison of token selection at 10% retention. For GAZE, gold and blue pixels denote tokens selected by the Focal and Spatial channels, respectively. For baselines, pink pixels indicate retained tokens.

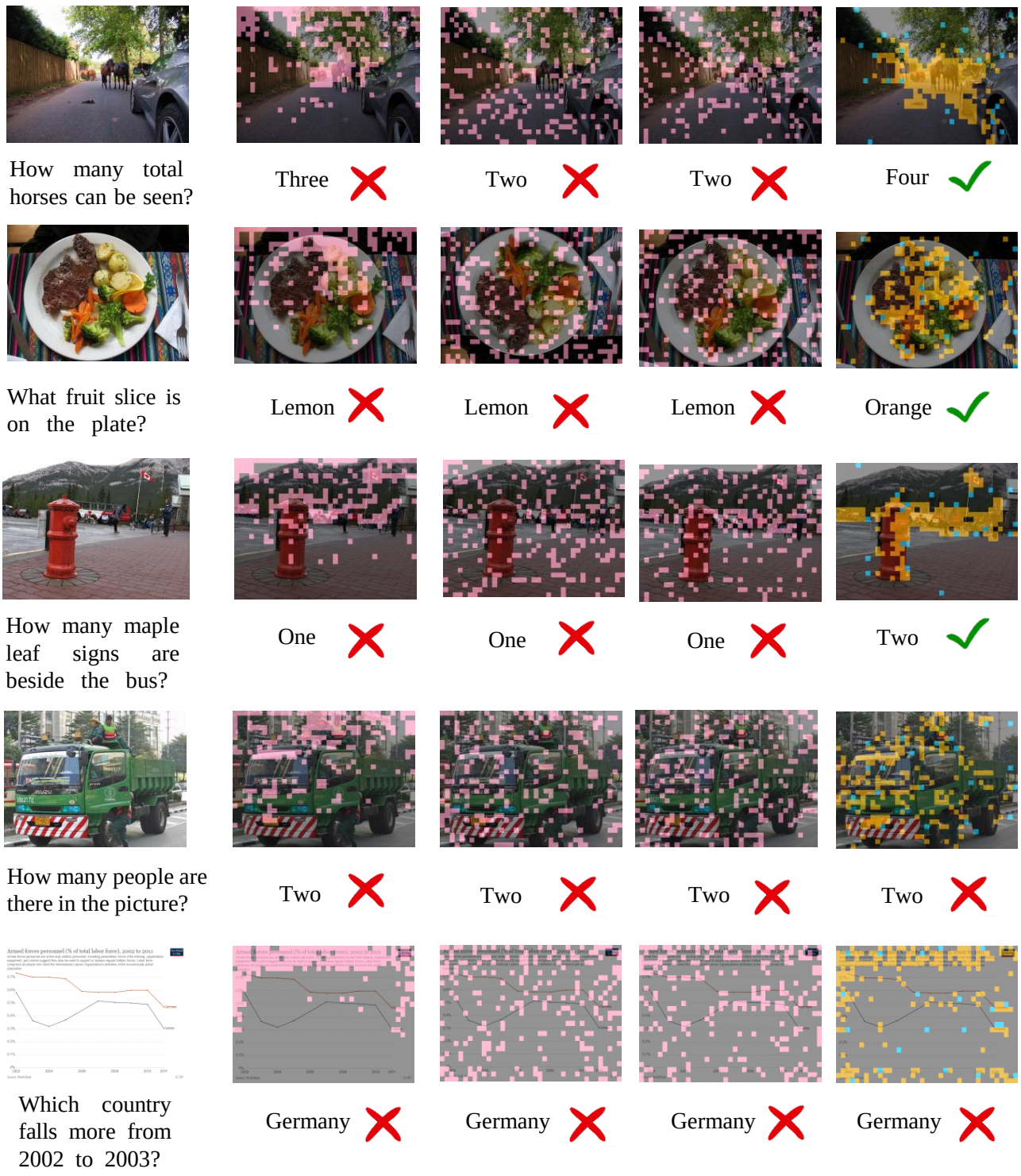


Figure 2: Qualitative comparison of token selection at 20% retention. For GAZE, gold and blue pixels denote tokens selected by the Focal and Spatial channels, respectively. For baselines, pink pixels indicate retained tokens.

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